TEJAS ROHIT KHARKAR

AI & Machine Learning Software Engineer | Computer Vision | Edge AI & TinyML Specialist

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PROFESSIONAL SUMMARY

Innovative AI & Machine Learning Software Engineer with 2+ years of experience architecting real-time Computer Vision pipelines, Generative AI RAG engines, Edge AI systems, and low-latency deep learning inference workflows. Demonstrated expertise in YOLOv11 object detection, MediaPipe 3D spatial vision, model quantization (INT8 TFLite Micro/ONNX), and full-stack REST API containerization using FastAPI and Docker. Recognized with 9 national and international honors across top institutions including IIIT Delhi, NIT Surathkal, COEP Pune, and Technoxian World Robotics Championship.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL, C, HTML5/CSS3 | **AI / ML & Vision:** PyTorch, TensorFlow, YOLOv11, Scikit-Learn, LightGBM, OpenCV, MediaPipe, CUDA | **GenAI & LLMs:** LangChain, LlamaIndex, ChromaDB, HuggingFace, RAG Architectures, BGE Embeddings | **Edge AI & Embedded:** TinyML, TensorFlow Lite Micro, ESP32-CAM, STM32, Embedded C/C++, NimBLE, ROS2 Humble | **Tools & DevOps:** FastAPI, Docker, Next.js, React, Git, Linux, PlatformIO, Streamlit, REST APIs

HONORS & KEY ACHIEVEMENTS

- **1st Winner** – ESYA '23 Innovation Challenge at IIIT Delhi (2023)
- **1st Winner** – Project Expo at MIT Jaipur (2023)
- **International Winner (Runner Up)** – Technoxian RC Electric World Robotics Championship (2024)
- **1st Runner Up** – Virtual Robotics, MindSpark 2023 at COEP Pune (2023)
- **1st Runner Up** – Makers Carnival Innovation Challenge at JECRC Jaipur (2023)
- **2nd Runner Up** – IEEE National Level Project Expo at Chandigarh University | **2nd Runner Up** – Engineer 2k23 at NIT Surathkal
- **2nd Place** – National Innovation Summit at NIT Srinagar | **3rd Place** – Makers Exhibition at JECRC Jaipur

WORK EXPERIENCE

Hashtee Labs – *AI Engineer* | Mumbai, India (May 2026 – Present)

- Engineered industrial Computer Vision and Edge AI surveillance pipelines by fine-tuning YOLOv11 architectures, resulting in real-time safety gear detection with 96.5% mAP@50 across multi-stream RTSP camera feeds.
- Accelerated video frame inference throughput by 42% through CUDA FP16 tensor core optimization and custom OpenCV multi-threaded frame buffers, lowering per-frame processing latency from 68ms to <40ms.
- Architected containerized microservice REST APIs using FastAPI and Docker, achieving 99.9% deployment uptime and scaling endpoint processing to 150+ concurrent requests/second.

Shunya – *Technical Lead – Embedded & Edge AI* | Jaipur / Remote (Aug 2023 – Present)

- Led hardware-software R&D for 5+ production TinyML prototypes, designing complete machine learning lifecycles from raw sensor data acquisition to INT8 model deployment on ESP32 microcontrollers.
- Reduced embedded model RAM footprint to <80KB SRAM by quantizing MobileNetV2 classifiers using TensorFlow Lite Micro, enabling on-device real-time visual inference at 48ms per frame without external DRAM.
- Engineered a BLE Smart Lock security system using NimBLE C++ and RSSI proximity authentication, establishing an AES-128 challenge-response cryptographic handshake that eliminated unauthorized access.
- Designed an active tremor cancellation algorithm for Parkinson's assistive devices by coupling MPU6050 6-DOF IMU complementary filters with adaptive PID servo loops, suppressing tremor amplitude by 91.4%.

SELECTED PROJECTS

AI PPE Safety Detection Pipeline | *PyTorch, YOLOv11, OpenCV, FastAPI, Docker* | GitHub Repo

- Accomplished 96.5% mAP@50 detection precision by implementing multi-threaded OpenCV video streams and PyTorch CUDA acceleration, resulting in a 75% reduction in manual safety audit overhead.

Enterprise GenAI RAG Knowledge Engine | *Llama 3 8B, ChromaDB, BGE, LangChain, FastAPI* | GitHub Repo

- Accelerated document retrieval speed by 65% by engineering vector similarity search and custom prompt memory chains, delivering sub-1.2s streaming query response times with zero external API dependencies.

Real-Time Driver Drowsiness & Fatigue AI | *MediaPipe, OpenCV, PyTorch, SciPy* | GitHub Repo

- Reduced false alarm rate to <1.5% by implementing consecutive frame threshold verification on 468 3D landmark meshes, triggering sub-second audio-visual warning alarms upon detecting fatigue events.

Edge AI Object Detection on ESP32-CAM (TinyML) | *TFLite Micro, C++, ESP32, INT8 Quantization* | GitHub Repo

- Achieved 48ms standalone edge inference latency by implementing INT8 quantized TFLite Micro tensor execution in 80KB SRAM, eliminating cloud communication costs entirely.

EDUCATION

Bachelor of Technology (B.Tech) in Artificial Intelligence – University of Engineering & Management (UEM), Jaipur (2023 – 2027)